How-To Library

Informatica

Reading Profiles in Third-Party Applications

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Abstract

Informatica provides database views that you can use to open profile results in third-party reporting tools. This article describes how to configure the database views for the profile statistics you want to read.

Supported Versions

- Informatica Data Explorer 9.0.1 HotFix 1 - 9.5.1

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Overview

When you run a profile in Informatica Analyst or Informatica Developer, the Data Integration Service stores the results in a dedicated database called the *profile warehouse*. When you open a profile in the Analyst tool or Developer tool, the Data Integration Service reads the results from the profile warehouse.

You can use third-party reporting tools to read profile results from the profile warehouse. Informatica provides a set of profile views that you can customize for the profile statistics you want to read. These views are based on common types of profile statistics and profile results analysis.

The product installer writes the profile views to the profiling warehouse. The views scripts take the form of SQL queries. Use any reporting tool that supports SQL queries and that provides a user interface for database views.

Profile View Definitions

When you install Informatica Data Explorer or Informatica Data Quality, the installation process adds view definitions to the profile warehouse.

The following table describes the view definitions:

Name	Description	Supported Version		
		9.0.1 HotFix 1 - 9.0.1 HotFix 2	9.1.0	9.5.0 or later
IDPV_BOTTOM_10_FREQUENCIES	The ten least frequent values for all fields in the profile.	Yes	Yes	Yes
IDPV_COL_PROFILE_RESULTS	The results summary of all column profile functions.	Yes	Yes	Yes
IDPV_DATA_DOMAIN_DETAILS	The latest data domain discovery results from the enterprise discovery results of a profile model.	No	No	Yes
IDPV_DATA_DOMAINS_GLOSSARY	The list of data domains in the data domain glossary and the data domain groups they belong to.	No	No	Yes
IDPV_DATA_DOMAINS_RESULTS	The latest data domain discovery results of a profile.	No	No	Yes

Name	Description	Supported Version		
		9.0.1 HotFix 1 - 9.0.1 HotFix 2	9.1.0	9.5.0 or later
IDPV_ENTITY_DETAILS	The primary key and foreign key relationships of data objects for each entity in the enterprise discovery results.	No	No	Yes
IDPV_ENTITY_VIEW	The latest information on related, self-related, and unrelated data objects in enterprise discovery results.	No	No	Yes
IDPV_PATTERN_FREQ_TRENDING	A trend of pattern inference based on multiple profile runs.	No	Yes	Yes
IDPV_PATTERN_INF_RESULTS	The result of pattern inference function.	Yes	Yes	Yes
IDPV_PROF_FDA_RESULTS	The functional dependency inference results.	No	Yes	Yes
IDPV_PROF_PK_RESULTS	The primary key inference results.	No	Yes	Yes
IDPV_PROFILE_DETAILS	The profile metadata summary.	Yes	Yes	Yes
IDPV_PROFILE_DETAILS_TRENDING	A trend of profile metadata information based on multiple profile runs.	No	Yes	Yes
IDPV_PROFILE_RESULTS_TRENDING	A trend of profile results based on multiple profile runs.	No	Yes	Yes
IDPV_SCORE_SMRY	The latest scorecard results.	No	No	Yes
IDPV_TOP_10_FREQUENCIES	The ten most frequent values for all fields in the profile.	Yes	Yes	Yes

Name	Description	Supported Version		
		9.0.1 HotFix 1 - 9.0.1 HotFix 2	9.1.0	9.5.0 or later
IDPV_VAL_FREQ_RESULTS	The result of value frequency function.	Yes	Yes	Yes
IDPV_VAL_FREQ_TRENDING	A trend of value frequency results based on multiple profile runs.	No	Yes	Yes

Defining the SQL Query

Use a Select statement to read the profile results in your reporting tool.

Create the Select statement with the following syntax:

```
Select * from [profile_view_name] where profile_id = 'PROFILE_ID'
```

Finding the Profile ID

To find the profile ID, complete the following steps:

1. Open a third-party reporting tool to query the profile warehouse.
2. Open the **IDPV_PROFILE_DETAILS_TRENDING** view.
3. The **PROFILE_ID** column displays the ID of the profile.

Viewing the Profile Results

Run the SQL query to view the profile results. All views return latest successful results for all profiles available in the profile warehouse. You can filter the results to a specific profile by querying the view with a Where clause.

To run the query, complete the following steps:

1. Open the reporting application.
2. Define a SQL query in the following format:
   For example, the following query reads the top ten most frequent values across all columns in a profile with ID U:P-7WcvxKEd6r6i0WujLkCA:

   Select * from : IDPV_TOP_10_FREQUENCIES where profile_id = 'U:P-7WcvxKEd6r6i0WUjLkCA'
3. Run the query and view the results.

Profile View Descriptions

Each profile view exposes different types of data within the profile warehouse. Choose the profile view that provides the information you need.

IDPV_BOTTOM_10_FREQUENCIES

This view exposes the ten least frequent values for all columns you ran the profile on.

The following table describes the columns in the IDPV_BOTTOM_10_FREQUENCIES view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
FIELD_NAME	Name of the data source column.
FIELD_VALUE	Value in the FIELD_NAME column.
FREQUENCY_COUNT	Number of occurrences of FIELD_VALUE in the FIELD_NAME column.
LAST_TIME_RUN	Date and time of the last profile run.

IDPV_COL_PROFILE_RESULTS

This view exposes the summary results of all column profile functions.

The following table describes the columns in the IDPV_COL_PROFILE_RESULTS view:

Column	Description
PROFILE_ID	ID of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
FIELD_NAME	Name of the data source column.
DOCUMENTED_DATATYPE	Datatype specified for the FIELD_NAME column during the data source import.
INFERRED_DATATYPE	Datatype inferred for the FIELD_NAME column by the Profiling Service Module. The Profiling Service Module is a component of the Data Integration Service.
MAXIMUM_VALUE	Maximum value in the FIELD_NAME column.
MINIMUM_VALUE	Minimum value in the FIELD_NAME column.
DISTINCT_VALUE	Number of unique values in the FIELD_NAME column.
DISTINCT_PERCENT	Number of unique values in the FIELD_NAME column expressed as a percentage of the number of rows you ran the profile on.
NULL_COUNT	Number of nulls in the FIELD_NAME column.

Column	Description
TOTAL_ROWS	Number of rows you ran the profile on.
LAST_TIME_RUN	Date and time of the last profile run.

IDPV_DATA_DOMAIN_DETAILS

This view exposes the latest data domain discovery results from the enterprise discovery results of a profile model.

The following table describes the columns in the IDPV_DATA_DOMAIN_DETAILS view:

Column	Description
PROFILE_MODEL_ID	ID of the profile model.
PROFILE_TASK_ID	ID of the profile task.
PROFILE_TASK_NAME	Name of the profile task.
PROFILEABLE_RECORD_ID	ID of the source in the profile definition.
PROFILEABLE_RECORD_NAME	Name of the source in the profile definition.
SOURCE_NAME	Name of the data source.
SOURCE_ID	ID of the data source.
CONNECTION_NAME	Name of the connection.
ROW_IDENTIFIER	System-generated row ID.
FIELD_NAME	Name of the data source column.
NATIVE_FIELD_TYPE	Specified datatype for the FIELD_NAME column for data source import.
DATA_DOMAIN_TYPE	Type of data domain inference.
DATA_DOMAIN_NAME	Name of the data domain.
SATISFIED_COUNT	The number of rows that match data domain conformance percentage.
SATISFIED_COUNT_PERCENT	The number of rows that match data domain conformance percentage, expressed as a percentage of the total number of rows.
NULL_COUNT	The number of null values in the data source.
NULL_COUNT_PERCENT	The number of null values in the data source, expressed as a percentage of the total number of rows.
TOTAL_ROWS	The total number of rows in the data source.
LAST_TIME_RUN	Date and time of the last profile model run.
DATA_DOMAIN_KEY	ID of the data domain.

IDPV_DATA_DOMAINS_GLOSSARY

This view exposes the data domains in the data domain glossary and the data domain groups they belong to.

The following table describes the columns in the IDPV_DATA_DOMAINS_GLOSSARY view:

Column	Description
DATA_DOMAIN_KEY	ID of the data domain.
DATA_DOMAIN_NAME	Name of the data domain.
DATA_DOMAIN_GROUP_NAME	Name of the data domain group the data domain belongs to.

IDPV_DATA_DOMAINS_RESULTS

This view exposes the latest data domain discovery results of a profile.

The following table describes the columns in the IDPV_DATA_DOMAINS_RESULTS view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
RECORD_ID	ID of the data source you ran the profile on.
FIELD_NAME	Name of the data source column.
DATA_DOMAIN_TYPE	Type of data domain inference.
DATA_DOMAIN_NAME	Name of the data domain.
SATISFIED_COUNT	The number of rows that match data domain conformance percentage.
SATISFIED_COUNT_PERCENT	The number of rows that match data domain conformance percentage, expressed as a percentage of the total number of rows.
NULL_COUNT	The number of null values in the data source.
NULL_COUNT_PERCENT	The number of null values in the data source, expressed as a percentage of the total number of rows.
TOTAL_ROWS	The total number of rows in the data source.
DATA_DOMAIN_KEY	ID of the data domain.

IDPV_ENTITY_DETAILS

This view exposes the details of primary key and foreign key relationships of data objects for each entity in the enterprise data discovery results.

The following table describes the columns in the IDPV_ENTITY_DETAILS view:

Column	Description
PROFILE_MODEL_ID	ID of the profile model.
PROFILE_TASK_NAME	Name of the profile task.
ENTITY_NAME	Name of the entity.
PROFILEABLE_RECORD_NAME	Name of the source in the profile definition.
PROFILEABLE_RECORD_ID	ID of the source in the profile definition.
SOURCE_NAME	Name of the data source.
SOURCE_ID	ID of the data source.
CONNECTION_NAME	Name of the connection.
ROW_IDENTIFIER	System-generated row ID.
RELATIONSHIP_TYPE	Type of relationship such as primary key to primary key or primay key to foreign key.
EST_PARENT_CONTAINMENT_PERCENT	Estimated percentage of parent data object rows in the child data object.
EST_CHILD_CONTAINMENT_PERCENT	Estimated percentage of child data object rows in the parent data object.
EST_PARENT_UNMATCHED_PERCENT	Estimated percentage of unmatched data in the parent data object.
EST_CHILD_UNMATCHED_PERCENT	Estimated percentage of unmatched data in the child data object.
EST_PARENT_NULL_PEECENT	Estimated percentage of null values in the parent data object.
EST_CHILD_NULL_PERCENT	Estimated percentage of null values in the child data object.
EST_OVERLAP_PERCENT	Estimated percentage of overlap between the paent and child data objects.
TOTAL_ROWS_IN_PARENT	Total numberof rows in the parent data object.
TOTAL_ROWS_IN_CHILD	Total numberof rows in the child data object.
SEQ_NUMBER	System-generated number for sequencing of rows.
FIELD_NAME	Name of the data source column.
FIELD_POSITION	The order of the field in a composite key when the Developer tool infers a relationship.

Column	Description
IS_SOURCE_PK	Indicates whether the columns in foreign key analysis are a part of primary keys.
LAST_TIME_RUN	The date and time of profile task run.

IDPV_ENTITY_VIEW

This view exposes the latest information on related, self-related, and unrelated data objects of enterprise data discovery results.

The following table describes the columns in the IDPV_ENTITY_VIEW view:

Column	Description
PROFILE_MODEL_ID	ID of the profile model.
ENTITY_NAME	Name of the entity.
ENTITY_TYPE	Type of the entity such as related, self-related, or unrelated data objects.
PROFILEABLE_RECORD_ID	ID of the source in the profile definition.
PROFILEABLE_RECORD_NAME	Name of the source in the profile definition.
SOURCE_NAME	Name of the data source.
SOURCE_ID	ID of the data source.
SOURCE_TYPE	Type of the data source such as a flat-file or relational data source.
ENTITY_ID	ID of the entity.
CONNECTION_NAME	Name of the connection.

IDPV_PATTERN_FREQ_TRENDING

This view exposes the pattern inference trends over multiple profile runs.

The following table describes the columns in the IDPV_PATTERN_FREQ_TRENDING view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
FIELD_NAME	Name of the data source column.
DOMAIN_VALUE	The pattern inferred for the FIELD_NAME column.
FREQUENCY	Frequency of the values matching each pattern in the FIELD_NAME column.

Column	Description
TOTAL_ROWS	Number of rows you ran the profile on.
FREQUENCY_PERCENT	Frequency of matching values as a percentage of the number of rows you ran the profile on.
LAST_TIME_RUN	Date and time of the last profile run.

IDPV_PATTERN_INF_RESULTS

This view exposes the results of pattern inference function.

The following table describes the columns in the IDPV_PATTERN_INF_RESULTS view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
FIELD_NAME	Name of the data source column.
DOMAIN_VALUE	The pattern inferred for the FIELD_NAME column.
FREQUENCY	Frequency of the values matching each pattern in the FIELD_NAME column.
FREQUENCY_PERCENT	Frequency of matching values as a percentage of the number of rows you ran the profile on.
TOTAL_ROWS	Number of rows you ran the profile on.
LAST_TIME_RUN	Date and time of the last profile run.

IDPV_PROF_FDA_RESULTS

This view exposes the functional dependency inference results.

The following table describes the columns in the IDPV_PROF_FDA_RESULTS view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
DETERMINANT	Name of the column that determines the values of the DEPENDENT_FIELD column.
DEPENDENT_FIELD	Name of the column containing values that are determined by the DETERMINANT column.
VALID_ROWS_PERCENT	Number of valid rows expressed as a percentage.

IDPV_PROF_PK_RESULTS

This view exposes the primary key inference results.

The following table describes the columns in the IDPV_PROF_PK_RESULTS view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
PRIMARY_KEY	Primary key value in the data source.
VALID_ROWS_PERCENT	Number of valid rows expressed as a percentage.

IDPV_PROFILE_DETAILS

This view exposes profile metadata information.

The following table describes the columns in the IDPV_PROFILE_DETAILS view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
FIELD_NAME	Name of the data source column.

IDPV_PROFILE_DETAILS_TRENDING

This view exposes the trend of profile metadata information over multiple profile runs.

The following table describes the columns in the IDPV_PROFILE_DETAILS_TRENDING view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
RULE_NAME	Name of the rule.
FIELD_NAME	Name of the data source column.
TIME_CREATED	Date and time of last profile run for the column.

IDPV_PROFILE_RESULTS_TRENDING

This view exposes the trend of profile results over multiple profile runs.

The following table describes the columns in the IDPV_PROFILE_RESULTS_TRENDING view:

Column	Description
PROFILE_ID	ID of the profile.
RECORD_NAME	Name of the data source you ran the profile on.
FIELD_NAME	Name of the data source column.
DISTINCT_VALUE	Number of unique values in the FIELD_NAME column.
NULL_COUNT	Number of null values in the FIELD_NAME column.
TOTAL_ROWS	Number of rows you ran the profile on.
TIME_CREATED	Date and time of last profile run for the column.

IDPV_SCORE_SMRY

This view exposes the latest scorecard results for each scorecard.

The following table describes the columns in the IDPV_SCORE_SMRY view:

Column	Description
SCORE_NAME	Name of the scorecard metric.
TOTAL_ROWS	Total number of rows in the data source.
INVALID_ROWS	Number of invalid rows for each scorecard column.
SCORE_PERCENTAGE	Score value expressed as a percentage of valid records.
RECORD_NAME	Name of the data source you ran the profile on.
FIELD_NAME	Name of the data source column.
TIME_CREATED	Date and time of the scorecard run.
SCORECARD_NAME	Name of the scorecard.
SCORECARD_IDENTIFIER	ID of the scorecard.

IDPV_TOP_10_FREQUENCIES

This view exposes the ten most frequent values for all columns you ran a profile on.

The following table describes the columns in the IDPV_TOP_10_FREQUENCIES view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
FIELD_NAME	Name of the data source column.
FIELD_VALUE	Value in the FIELD_NAME column.
FREQUENCY_COUNT	Number of occurrences of the FIELD_VALUE in the FIELD_NAME column.
LAST_TIME_RUN	Date and time of the last profile run.

IDPV_VAL_FREQ_RESULTS

This view exposes the results of value frequency function.

The following table describes the columns in the IDPV_VAL_FREQ_RESULTS view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.
FIELD_NAME	Name of the data source column.
FIELD_VALUE	Value in the FIELD_NAME column.
FREQUENCY_COUNT	Number of occurrences of the FIELD_VALUE in the FIELD_NAME column.
FREQUENCY_PERCENT	The FREQUENCY_COUNT value expressed as a percentage of the number of rows you ran a profile on.
LAST_TIME_RUN	Date and time of the last profile run.

IDPV_VAL_FREQ_TRENDING

This view exposes the trend of value frequency results over multiple profile runs.

The following table describes the columns in the IDPV_VAL_FREQ_TRENDING view:

Column	Description
PROFILE_ID	ID of the profile.
PROFILE_NAME	Name of the profile.

Column	Description
FIELD_NAME	Name of the data source column.
FIELD_VALUE	Value in the FIELD_NAME column.
FREQUENCY_COUNT	Number of occurrences of the FIELD_VALUE in the FIELD_NAME column.
FREQUENCY_PERCENT	The FREQUENCY_COUNT value expressed as a percentage of the number of rows you ran the profile on.
LAST_TIME_RUN	Date and time of the last profile run for the column.

Conclusion

You can use Informatica Analyst and Informatica Developer to create and run profiles that analyze the structure and content of data. You can view profile results, build data scorecards, and drill down to the underlying data in the Analyst tool and Developer tool.

You can also view profile data in third-party reporting applications. Use the SQL queries described in this article to view profile data in third-party applications.

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